

Mauro Malekshamran

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EDUCATION

San Diego State University

May 2027 (Expected)

Master of Science in Computer Science

California State University of San Marcos

May 2025

Bachelor of Science in Computer Science

EXPERIENCE

Research Assistant, Persistent Memory Systems

Mar 2026 - Current

San Diego State University

San Diego, CA

- Conduct research on persistent memory data structures, focusing on spiral storage and dynamic hashing techniques to improve performance and scalability for large scale key value systems
- Analyze tradeoffs between concurrency, caching strategies, and memory allocation to understand differences between persistent memory and traditional DRAM based systems
- Explore collision handling and lookup efficiency in hash based storage, evaluating probing methods, mapping strategies, and system level optimizations in C++
- Reviewed and extended existing research papers, identifying limitations and proposing improvements in storage design and access patterns
- Collaborated with faculty through technical discussions and iterative feedback, strengthening understanding of system architecture, memory behavior, and performance bottlenecks

Computer Science & Mathematics Tutor

Aug 2024 - May 2025

CSUSM STEM Success Center

San Marcos, CA

- Tutored 150+ students one-on-one in C++, Python, MIPS Assembly delivering over 200 hours of personalized sessions on data structures, algorithms, and debugging, driving an average 35% grade improvement
- Organized 10+ targeted C++ and Calculus workshops for 20+ students, authoring practice exams and reviewing 400+ assignments, boosting average exam scores by 25%
- Collaborated with faculty and peers to identify common learning gaps, implementing focused review sessions that helped over 85% of attending students improve at least one letter grade in STEM courses

Coding Instructor

Jan 2023 - Aug 2024

Rock Creek Education

Vista, CA

- Instructed 120+ elementary and middle school students in Python and C++ through 150+ hours of interactive, STEM-aligned curriculum, strengthening core problem-solving and computational thinking skills
- Designed and facilitated 15+ project-based learning workshops where students developed 100+ functional coding projects

PROJECTS

Historical Figures Voice Assistant | Python, Qwen, LoRA, PostgreSQL, Tkinter, Pandas, Transformers

Feb 2026 - May 2026

- Developed a voice enabled historical figures assistant that transcribes spoken queries and routes them into a **Qwen** based response **pipeline**
- Prepared and structured data from a **PostgreSQL** knowledge base containing **11K plus** biographies, historical events, and QA pairs for training and evaluation
- Built a lightweight cross platform **Tkinter** GUI for audio capture, transcription display, and response output, with a deployment path for both desktop and **Raspberry Pi** environments
- Collaborated on **LoRA** based model adaptation and Raspberry Pi plus **Hailo** deployment by supporting the interface, transcription, and evaluation components of the system

Drone Detection System | Python, PyTorch, YOLOv5, OpenCV, CUDA, 4K Imaging

Sep 2025 - Dec 2025

- Designed and trained a **YOLO** based drone detection pipeline using high resolution 4K imagery and multi-source datasets including **Kaggle** and **HuggingFace**, achieving **~0.90 mAP@0.5** on controlled test data
- Implemented fog augmentation and domain adaptation techniques to improve robustness under low visibility conditions, increasing real world generalization despite reduced training epochs
- Evaluated **model performance** using mAP, F1 score, precision recall curves, confusion matrices, and loss convergence

TECHNICAL SKILLS

Languages: Python, C/C++, JavaScript, TypeScript, SQL, MIPS Assembly

Frameworks & Libraries: TensorFlow, Keras, PyTorch, scikit-learn, TFLite, Pandas, NumPy, Flask, OpenCV, Yolo

Tools & Services: GitHub, Linux/Unix, AWS, Azure, CI/CD tools, Docker, REST APIs, VS Code, Unit testing

Databases: MySQL, PostgreSQL, Firebase, Supabase, Data management, Backend/Frontend

Machine Learning & AI: Reinforcement Learning, Computer Vision, Model Validation/Testing, RAG, Yolo, Linear Algebra

Experience With: LLM/LLVM, CNNs, RNNs, XLA, NLPs, Deep learning, Go